

Shreesha G. Bhat

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RESEARCH INTERESTS

Distributed Systems, Data Streaming, Systems for Agentic AI, Machine Learning Systems, Storage Systems, CXL

EDUCATION

University of Illinois Urbana-Champaign 2023 - 2028 (expected)
PhD in Computer Science | Advisors: [Ram Alagappan](#), [Aishwarya Ganesan](#) | CGPA: 4.0/4.0

Indian Institute of Technology, Madras 2018 - 2023
Dual Degree (BTech + MTech) in Computer Science & Engineering | Advisor: [Kartik Nagar](#) | CGPA: 9.66/10.00

PUBLICATIONS

- SOSP '26** [AgileLog: A Forkable Shared Log for Agents on Streaming Data](#)
[Shreesha G. Bhat, Tony Hong, Michael Noguera, Aishwarya Ganesan, Ramnatthan Alagappan](#)
The 32nd Symposium on Operating Systems Principles (SOSP) 2026 (to appear)
- OSDI '26** [MEGALON: Efficient Data Sharing for Partly Coherent CXL Memory](#)
[Jiyu Hu, Seokjoo Cho*, Landon Johnson*, Kiran Hombal, Shreesha G. Bhat, Marcos K. Aguilera, Ramnatthan Alagappan, Aishwarya Ganesan \(*equal contribution\)](#)
The 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI) 2026
Jay Lepreau Best Paper Award Nominee
- EuroSys '26** [A Logically Disaggregated Cache for Replicated Storage Systems](#)
[Kiran Hombal, Henry Zhu, Shreesha G. Bhat, Neil Kaushikkar, Ramnatthan Alagappan, Aishwarya Ganesan](#)
The 21st European Conference on Computer Systems (EuroSys) 2026
- OSDI '25** [Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering](#)
[Shreesha G. Bhat, Tony Hong, Xuhao Luo, Jiyu Hu, Aishwarya Ganesan, Ramnatthan Alagappan](#)
The 19th USENIX Symposium on Operating Systems Design and Implementation (OSDI) 2025
Covered on the Metadata Blog
- SOSP '24** [LazyLog: A New Shared Log Abstraction for Low-Latency Applications](#)
[Xuhao Luo, Shreesha G. Bhat*, Jiyu Hu*, Ramnatthan Alagappan, Aishwarya Ganesan \(*equal contribution\)](#)
The 30th Symposium on Operating Systems Principles (SOSP) 2024
Best Paper Award | Invited to Transactions on Computer Systems | Inspired Lightning Topics in Warpstream
- FMCAD '23** [Automating Cutoff-based Verification of Distributed Protocols](#)
[Shreesha G. Bhat, Kartik Nagar](#)
Formal Methods in Computer-Aided Design (FMCAD) 2023
- DISC '21** [Brief Announcement: Automating and Mechanising Cutoff Proofs for Parameterized Verification of Distributed Protocols](#)
[Shreesha G. Bhat, Kartik Nagar](#)
35th International Symposium on Distributed Computing (DISC) 2021

WORKSHOP PUBLICATIONS

- CAIS '26
SAO** [Agents for Data Streaming Tasks: The Missing Pieces](#)
[Shreesha G. Bhat*, Landon Johnson*, Michael Noguera*, Aishwarya Ganesan, Ramnatthan Alagappan \(*equal contribution\)](#)
Supporting our AI Overlords Workshop at CAIS 2026 (SAO) May 2026
Spotlight

REFEREED JOURNAL PUBLICATIONS

- TOCS '25** [LazyLog: A New Shared Log Abstraction and Design for Modern Low-Latency Applications](#)
[Xuhao Luo, Shreesha G. Bhat*, Jiyu Hu*, Ramnatthan Alagappan, Aishwarya Ganesan \(*equal contribution\)](#)
ACM Transactions on Computer Systems (TOCS) August 2025
Invited for Publication

RESEARCH EXPERIENCE

Research @ Distributed and Storage Systems Lab | UIUC
Research Assistant | Guide: [Ram Alagappan](#), [Aishwarya Ganesan](#)

Aug '23 – present

- Designed and implemented multiple shared log abstractions and systems for modern application paradigms.
- Contributed to projects on CXL data sharing and replicated storage systems.

Networking and Distributed Systems Lab | Hewlett Packard Labs

May'25 – August'25

Research Associate Intern | Guide: [Puneet Sharma](#)

- Designing infrastructure for agentic AI applications.

Improving Cloud Reliability through Systematic Testing | Microsoft Research India

Aug '22 – Jul '23

Research Intern | Guide: [Akash Lal](#)

- Worked on improving reliability of Azure Cloud Services using concurrency testing tools such as [Coyote](#) for C++ programs.
- Built a deterministic concurrency testing framework for a production scale replication library, [Azure RSL](#), which provides an implementation of the Paxos consensus algorithm. Implemented several optimizations to improve state-space coverage.

Parameterized Verification of Distributed Protocols | IIT Madras

Sep '20 – Jul '23

Young Research Fellow | Guide: [Kartik Nagar](#)

- Investigated cutoff-based techniques for verifying that distributed protocols meet their specification irrespective of the size of the parameter they are instantiated with (such as number of nodes).
- Proposed a framework to mechanize simulation based proofs for cutoffs and applied the approach on a variety of distributed protocols using Z3 as a backend SMT solver.

AWARDS & ACHIEVEMENTS

- Awarded the **Best Paper Award** at SOSP 2024 for the paper titled *LazyLog: A New Shared Log Abstraction for Low-Latency Applications*.
- Selected as a undergraduate research fellow as part of the (YRF) program.
- Secured prizes for excellent academic performance in the 1st, 2nd, 7th and 8th semesters at IIT Madras.
- Secured All India Rank of **851** and **619** in JEE (Joint Entrance Examination) Advanced & Mains 2018.
- Qualified for KVPY fellowship with an All India Rank of **142**.
- Among the **top 300** in India qualified to write national olympiad examinations for Physics, Chemistry and Astronomy (INPhO, INChO, INAO).

SERVICE

- **Artifact Evaluation Committees:** OSDI'24, ATC'24, FAST'26 (Spring and Fall)
- **Shadow Program Committees:** EuroSys'26
- **Student Volunteer:** SOSP'24

TEACHING

- **CS598 Storage Systems, TA, Fall 2025:** Held office hours, graded assignments, and led lecture sessions.

INVITED TALKS

- **Forkable Shared Logs: Enabling Testing, Analysis, and Agentic Workloads on Real-Time Data**

Stream Processing with Apache Kafka, Samza, and Flink Meetup hosted by LinkedIn

June 24, 2026

Location: Sunnyvale, CA + online

Links: [Meetup Event](#) | YouTube (forthcoming)

GRANTS

- **Conference Travel Grants:** OSDI'25, SOSP'24